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PROJECT TITLE

Predicting At-Risk Student Academic Performance Using Machine Learning: A Comparative Study of Supervised Classification Models

SUBMITTED TO:
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CHAPTER ONE

Introduction

This report presents an end-to-end supervised machine learning project for predicting students who may be at risk of failing a course. The work responds to the COEN807 term project requirement by selecting a credible public dataset, defining a practical learning problem, building reproducible models, comparing algorithms, and interpreting the results for real educational decision support.

Background of the Study

Machine learning is increasingly used in educational analytics because schools and universities need earlier evidence about students who may require academic support. A model that identifies risk before final results can help advisers, lecturers, and departments provide tutoring, counselling, or closer follow-up. However, such a model must be evaluated carefully because high accuracy alone can hide poor performance on the minority group of students who are actually at risk.

Problem Statement

The problem addressed in this project is the prediction of whether a student will be at risk of failing based on available demographic, school, family, behavioural, and academic attributes. The original final grade is converted into a binary target, where At_Risk equals 1 if the final grade G3 is less than 10 out of 20 and 0 otherwise.

Aim and Objectives

The aim of the study is to design and evaluate supervised classification models for predicting at-risk student academic performance. The specific objectives are stated below.

- To formulate the prediction task as a binary supervised learning problem.
- To justify the selected public dataset and describe its quality and limitations.
- To preprocess the data and engineer a clear At_Risk target variable.
- To compare several classification models using the same experimental protocol.
- To evaluate the models using recall, precision, F1-score, ROC-AUC, balanced accuracy, and confusion matrices.
- To discuss bias, ethical use, reproducibility, and practical deployment issues.

Scope and Significance

The scope is limited to the UCI Student Performance dataset and a binary classification task. Two experimental settings are considered: an early-warning setting that excludes prior period grades G1 and G2, and a performance-aware setting that includes them. The significance of the work is that it demonstrates how machine learning can support educational intervention while also showing why careful metric selection and ethical safeguards are necessary.

CHAPTER TWO

Literature Review

Supervised machine learning is concerned with learning a relationship between input features and known target labels. In classification problems, the target is categorical. For this study, the label indicates whether a student is at risk or not at risk. Educational prediction studies often use demographic information, attendance, study time, previous performance, and family background to estimate academic outcomes.

Supervised Classification

Classification algorithms learn decision boundaries that separate classes. Logistic Regression provides an interpretable linear baseline, Decision Tree models provide rule-based decisions, Random Forest and Gradient Boosting improve performance through ensembles of trees, and Support Vector Machine models search for a separating margin that can work well in high-dimensional feature spaces.

Evaluation of Imbalanced Data

The selected target is imbalanced because far fewer students are labelled as at risk than not at risk. In such cases, accuracy can be misleading. A model may obtain high accuracy by predicting the majority class but still fail to identify students who need support. For this reason, recall and F1-score for the At_Risk class are treated as important evaluation measures.

Dataset Description and Justification

The dataset was sourced from the UCI Machine Learning Repository. It contains student achievement records collected from two Portuguese secondary schools. The Portuguese language subset was selected because it contains 649 records, which is larger than the Mathematics subset and is suitable for a compact but meaningful supervised learning experiment.

Dataset property	Value
Source	UCI Machine Learning Repository - Student Performance dataset
Selected file	student-por.csv
Records	649 students
Original variables	33
Target variable	At_Risk = 1 if G3 < 10, otherwise 0
Class distribution	100 at-risk and 549 not-at-risk students
Missing values	0 missing values in the selected file

CHAPTER THREE

Methodology

The study followed a reproducible machine learning workflow. The raw student dataset was loaded, the At_Risk target was engineered from the final grade, preprocessing was applied through a pipeline, several models were trained, and the best models were selected using cross-validation on the training data.

Data Preprocessing

Categorical features were encoded using one-hot encoding, while numerical features were imputed and standardised. Although the selected dataset had no missing values, imputation was included in the pipeline to make the workflow robust. The train-test split used 80% of the data for training and 20% for testing, with stratification to preserve the class ratio.

Feature Engineering

The final grade G3 was converted into the target variable At_Risk. A student was labelled as at risk when G3 was below 10. Two feature settings were designed. The early-warning experiment removed G1 and G2 so that the model used information available before later performance records. The performance-aware experiment included G1 and G2 to measure the improvement obtained when prior grades are available.

Model Design

Model	Reason for inclusion
Dummy Majority	Baseline for checking whether accuracy is misleading.
Logistic Regression	Interpretable linear classifier.
Decision Tree	Simple rule-based nonlinear model.
Random Forest	Robust ensemble of decision trees.
Gradient Boosting	Sequential ensemble that reduces errors iteratively.
Support Vector Machine	Margin-based classifier for nonlinear separation.

Experimental Design

Each model was trained under the two feature settings. Hyperparameters were tuned using five-fold stratified cross-validation on the training data only. The tuning score was F1-score for the At_Risk class because the project aims to detect vulnerable students while still controlling false alarms. Final results were computed once on the untouched test set, using accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix counts.

CHAPTER FOUR

Results and Discussion

The two experiments produced different practical insights. The early-warning experiment is harder because it excludes prior grades, but it is useful when intervention must begin before later academic records are available. The performance-aware experiment is more predictive because it includes G1 and G2, but it may identify risk later.

Early-Warning Experiment Without G1 and G2

In the early-warning setting, the Random Forest model produced the best At_Risk F1-score. It achieved 0.800 recall and 0.582 F1-score for at-risk students, meaning it identified 16 out of 20 at-risk students in the test set. The Dummy Majority baseline achieved high accuracy but detected no at-risk students, confirming that accuracy alone is not sufficient.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.823	0.457	0.800	0.582	0.822
Support Vector Machine	0.838	0.483	0.700	0.571	0.830
Decision Tree	0.738	0.325	0.650	0.433	0.670
Logistic Regression	0.777	0.355	0.550	0.431	0.794
Gradient Boosting	0.831	0.400	0.200	0.267	0.789
Dummy Majority	0.846	0.000	0.000	0.000	0.500

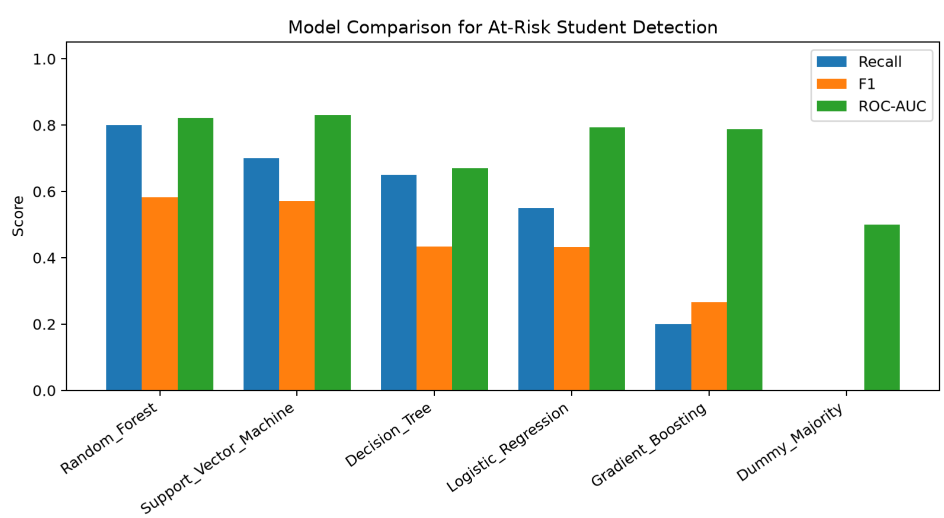

Figure 1: Early-warning model comparison.

Performance-Aware Experiment With G1 and G2

When G1 and G2 were included, overall predictive performance improved. The Support Vector Machine achieved the best At_Risk F1-score, with 0.908 accuracy, 0.850 recall, 0.739 F1-score, and 0.936 ROC-AUC. This shows that prior grades are strong predictors of final academic risk.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Support Vector Machine	0.908	0.654	0.850	0.739	0.935
Decision Tree	0.908	0.682	0.750	0.714	0.843
Random Forest	0.885	0.586	0.850	0.694	0.952
Logistic Regression	0.877	0.562	0.900	0.692	0.937
Gradient Boosting	0.892	0.625	0.750	0.682	0.931
Dummy Majority	0.846	0.000	0.000	0.000	0.500

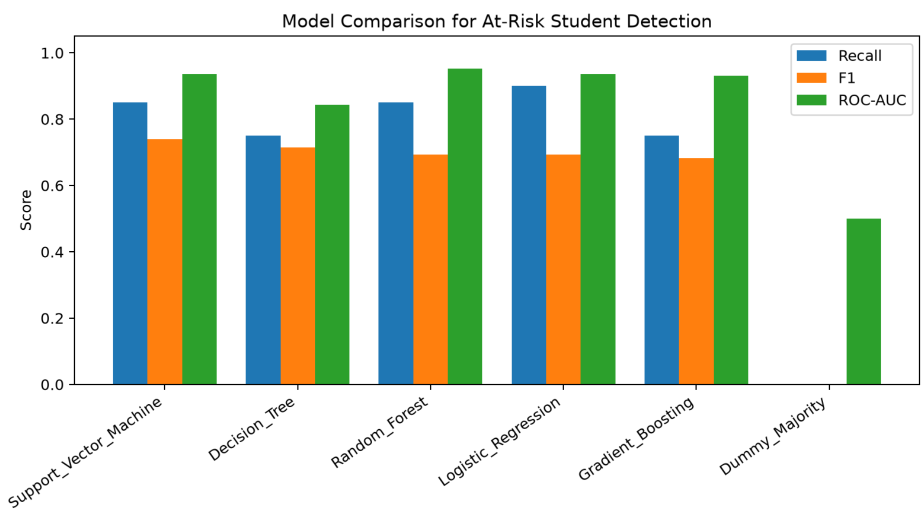

Figure 2: Performance-aware model comparison.

Critical Analysis

The main finding is that the best model depends on the intended use. If the aim is early intervention, the Random Forest early-warning model is useful because it does not depend on prior grades. If the aim is more accurate prediction after some academic records exist, the Support Vector Machine model is stronger. In both cases, the model should support human academic advising rather than replace it.

Bias and Ethical Considerations

The dataset contains variables such as address type, parental education, internet access, family support, and school support. These variables may reflect socioeconomic differences. Therefore, the model should guide support, not punishment or exclusion.

CHAPTER FIVE

Conclusion and Recommendations

This project demonstrates a complete supervised machine learning workflow for predicting at-risk student performance using a credible public dataset. The work covered problem formulation, dataset justification, preprocessing, model design, experimental comparison, evaluation, interpretation, ethical discussion, and reproducibility documentation.

Conclusion

The early-warning Random Forest model detected 80% of at-risk students without using prior grade variables, while the performance-aware Support Vector Machine achieved stronger overall performance after including G1 and G2. The results show that machine learning can support academic intervention, but the model must be evaluated using minority-class metrics and interpreted with caution.

Recommendations

- Validate the model on local Nigerian educational data before practical use.
- Use the model only as a support tool for tutoring, counselling, and academic advising.
- Monitor subgroup performance to reduce bias against disadvantaged students.
- Consider threshold tuning based on the number of students that support staff can realistically assist.
- Extend future work to include calibration analysis, feature importance, and additional classification algorithms.

Reproducibility and Repository

The project repository contains the source code, dataset/access instructions, processed data, trained models, result tables, visualizations, final report, presentation slides, requirements file, and reproduction instructions. The GitHub repository link is:
https://github.com/MusbahuTijjani/COEN807_Student_Performance_ML

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